

24. A non-renewable source of energy is :

- (a) Solar energy
- (b) Petroleum
- (c) Wind energy
- (d) Biogas

U.P.P.C.S. (Pre) 2021

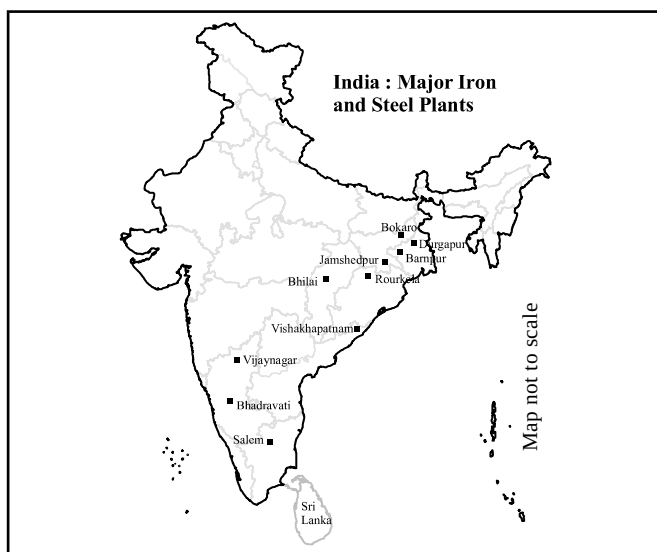
Ans. (b)

Petroleum, also called crude oil, is a **fossil fuel**. Like coal and natural gas, petroleum was formed from the remains of ancient marine organisms, such as plants, algae, and bacteria.

Industry

i. Iron and Steel Industry

*The iron and steel industry is a basic industry. The first attempt to make iron in India started in **1830 A.D.** in Porto Novo (Tamil Nadu), but this factory late failed. For the first time in India, iron was successfully produced by the Bengal Ironworks in **Kulti** in **1875 AD**. In the year 1907, a new factory was established in India by the Tata Iron and Steel Company at Sakchi (Jamshedpur). In where raw iron was produced in the year 1908 and steel was produced in the year 1912. According to World Steel Association data (World Steel in figures), 2024, it is the world's 10th largest (in the year, 2023) steel company (Tata in Steel Group, 2023) with an annual production capacity of 29.50 million tonnes.



***Visvesvaraya Iron and Steel Ltd. (VISL)** was established in the year 1923 on the bank of the Bhadravati

River in Shimoga District of Karnataka. Raw Iron is supplied from the **Bababudan Hills**, while limestone is sourced from Bhandigunda. Due to the lack of local supply of coal supply, charcoal or wood was used for power which was sourced from the **Malnad** forest area. Since 1989, it has acted as a subsidiary of the Steel Authority of India Ltd. It was completely merged with SAIL in the year 1998. *After the independence, during the second five-year plan of India, special emphasis was given to heavy and basic industries under which **Rourkela** (Odisha), **Bhilai** (Chhattisgarh), and **Durgapur** (West Bengal) steel plants were established.

***Rourkela Steel Plant or Hindustan Steel Limited (HSL)**, is located in the Sundargarh District of Odisha. This factory was established in collaboration with Germany's companies Krups and Demag. It is situated near deposits of manganese and limestone. Iron ore is brought from the Keonjhar and Kendujhar districts of Odisha. The cooking coal is obtained from Bokaro, Jharia and the Talcher area of the coal washing station in Kargali. Manganese, limestone and dolomite are obtained from Veermitrapur. It gets cheap hydroelectricity from Hirakud easily. ***The Bhilai steel plant** was established with the help of the former Soviet Union in Bhilai of Durg District in the state of Chhattisgarh on the Kolkata-Mumbai railroad. The **Dhallirajhara iron ore deposit** supplies iron ore to the Bhilai steel plant.

* **Hindustan Steel Limited (HSL)-Durgapur-** the Durgapur steel plant was established with the collaboration of Britain near Asansol in the Bardhaman district of West Bengal. This plant obtains iron ore from the Bolni mines of the Gua area, Coal from the mines of Barakar and Jharia, manganese from Jamda (Keonjhar District) and electricity and water from the Damodar River.

* **Bokaro Steel Limited (BSL) - Bokaro** -This plant was established with the collaboration of the then Soviet Union at the confluence of the Bokaro and Damodar rivers in the Hazaribagh district of the state of Jharkhand.

*Major minerals required for the iron and steel industry are iron ore, coking coal, manganese, rock phosphate, dolomite and chromium etc. Due to a lack of coking coal in the country, it is imported from Australia.

*Generally chromium and nickel are used in the making of stainless steel. Both of these are mixed with iron metal along with some quantity of carbon for the manufacturing of stainless-steel.

*The Visakhapatnam steel plant is India's first steel plant located on a coastline steel plant. Iron ore is supplied to this plant from Bailadila (Chhattisgarh).

1. What is used for manufacturing stainless steel?

- (a) Chromium & Nickel (b) Nickel & Copper
(c) Chromium Graphite (d) Benzene & Acetylene

U.P.P.C.S. (Pre) 1990

Ans. (a)

Stainless steel is a metal alloy, made up of steel mixed with elements such as chromium, nickel, molybdenum, silicon, aluminium and carbon. Iron mixed with carbon to produce steel is the main component of stainless steel. Hence, option (a) is the correct answer.

2. Stainless steel is an alloy of –

- (a) Iron and Copper (b) Iron and Zinc
(c) Iron and Chromium (d) Iron and Graphite

38th B.P.S.C. (Pre) 1992

M.P.P.C.S. (Pre) 2008

Ans. (c)

See the explanation of the above question.

3. The important metal used with iron to produce stainless steel is :

- (a) Aluminium (b) Chromium
(c) Tin (d) Carbon

U.P.P.C.S. (Pre) 2006

U.P.P.C.S. (Pre) 1997

Ans. (b)

See the explanation of the above question.

4. Some iron and steel plants have been planned along the western coast of India. What is the major reason for this locational shift in this industry?

- (a) Increased nuclear power generation in the Western Coastal region
(b) The occurrence of high-grade iron ore deposits in Goa and parts of Madhya Pradesh and the comparative ease of exporting steel from here.
(c) The decline in international demand for Indian iron ore from the Western coastal region.
(d) Adoption of sponge iron technology.

47th B.P.S.C. (Pre) 2005

Ans. (b)

Some iron and steel plants have been planned along the western coast of India. The major reasons for the locational shift of this industry are the occurrence of high-grade iron ore deposits in Goa and parts of Madhya Pradesh as well as sea-ports for comparative ease of exporting steel and consumption of iron ore in the concerned field for manufacturing.

5. In India, the steel production industry requires the import of

- (a) Saltpetre
(b) Rock phosphate
(c) Coking coal
(d) All of the above

I.A.S. (Pre) 2015

Ans. (c)

The iron and steel industry is one of the important industries in India. It requires minerals like iron ore, coking coal, manganese, dolomite, rock phosphate, chromium, etc. In India, coking coal is mainly imported from Australia. Coke is produced by heating coking coal. Coking coal is also known as metallurgical coal.

6. The local supply of coal is not available to :

- (a) TISCO, Jamshedpur
(b) VSL, Bhadravati
(c) HSL, Durgapur
(d) HSL, Bhilai

I.A.S. (Pre) 1996

Ans. (b)

VSL, Bhadravati is the only iron and steel plant situated outside the coal producing area. The absence of coal was met by the use of charcoal derived from firewood obtained from states forest. The plant switched over to electric furnaces after the completion of the Mahatma Gandhi Hydel Power Station.

7. Which one of the following iron and steel plant is located neither on iron ore area nor on coalfield but approximately at the midway between them?

- (a) Jamshedpur (b) Bhadravati
(c) Durgapur (d) Bokaro

Chhattisgarh P.C.S. (Pre) 2018

Ans. (a)

The Jamshedpur iron and steel plant is located almost in the middle of the two, not on the iron ore and coal fields.

8. TISCO plant is located near –

- (a) Patna (b) Darbhanga
(c) Dhanbad (d) Tatanagar

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

TISCO (Tata Iron & Steel Company Limited) is a prime steel company in India. It was established in 1907.

9. Pittsburgh of India is :

- (a) Bhagalpur (b) Varanasi

(c) Sindri

(d) Jamshedpur

Uttarakhand P.C.S. (Pre) 2021

Ans. (d)

Jamshedpur (Jharkhand) is known as the Pittsburgh of India. It is famous for its iron and steel industries.

10. Which one of the following statements is not correct?

- (a) Rourkela Steel Plant, the first integrated steel plant in the public sector of India was set up with the Soviet Union collaboration.
- (b) Salem Steel Plant is a premier producer of stainless steel in India.
- (c) Maharashtra Elektros melt Ltd. is a subsidiary of the Steel Authority of India Ltd.
- (d) Visakhapatnam Steel Plant is a unit of the Rashtriya Ispat Nigam Ltd.

I.A.S. (Pre) 2005

Ans. (a)

Rourkela Steel Plant, the first integrated steel plant in the public sector of India, was set up with the collaboration of Germany, not the Soviet Union. Hence, statement (a) is wrong. The other statements mentioned in the options are correct.

11. Bhilai Steel Plant is a collaborative project of the Indian government and :

- (a) Russia
- (b) Britain
- (c) Germany
- (d) Poland

U.P.P.C.S. (Mains) 2006

Ans. (a)

Bhilai Steel Plant is the collaborative project of the Indian government and Russia, located in Bhilai, Durg district in Chhattisgarh. The plant was commissioned in 1959. Presently, the total requirement of iron ore for Bhilai Steel Plant is met from Dalli Rajhara Iron Ore Complex (IOC), Korba and Kargali coal mines.

12. Bokaro Steel Plant was constructed with the collaboration of –

- (a) USA
- (b) Germany
- (c) England
- (d) Russia

Jharkhand P.C.S. (Pre) 2016

Ans. (d)

Bokaro Steel Plant is the fourth public sector steel plant in India built with Soviet (Russian) help. It was started in 1965. It was incorporated as a limited company on 29 January, 1964 and was later merged with SAIL (Steel Authority of India Limited). This plant is regarded as the country's first Swadeshi Steel Plant.

13. Which one of the following industries earns maximum foreign exchange for India :

- (a) Tea
- (b) Jute
- (c) Iron and steel
- (d) Sugar

U.P.P.C.S. (Mains) 2003

Ans. (c)

When the question was asked option (c) was the correct answer. According to data from the Ministry of Commerce and Industry, iron and steel exports have been significant in India's export basket. In contrast, while tea, jute and sugar are important agricultural exports for India, their individual contributions to total export earnings are comparatively lower than that of iron and steel. Therefore, among the given options, the iron and steel industry earns the maximum foreign exchange for India.

14. Which one of the following is not correctly matched?

- (a) Bhilai : Madhya Pradesh
- (b) Durgapur : West Bengal
- (c) Jamshedpur : Jharkhand
- (d) Rourkela : Odisha

U.P.P.C.S. (Spl) (Mains) 2004

Ans. (a)

Bhilai Steel Plant has been established with the help of the former Soviet Union. Durgapur Steel Plant (West Bengal) was set up with the help of the United Kingdom, while Rourkela Steel Plant (Odisha) was set up with the help of Germany. Jamshedpur (Jharkhand) steel plant belongs to the private sector.

15. Rourkela Steel Plant gets its supplies of iron ore from

- (a) Keonjhar
- (b) Dalli-Rajhara
- (c) Kamangundi
- (d) Mayurbhanj

U.P.P.C.S. (Mains) 2005

Ans. (a)